ABU NASIR PATWARY

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EDUCATION

United International University

B.Sc. in Computer Science and Engineering | CGPA: 3.22/4.00

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, JavaScript, PHP

Frameworks & Libraries: Django, Django REST Framework, Spring Boot, React.js, Tailwind CSS

API Technologies: REST APIs

Databases: MySQL, PostgreSQL

Machine Learning & Deep Learning: scikit-learn, PyTorch, Torchvision, timm

Computer Vision & Image Processing: OpenCV, Pillow, scikit-image

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Plotly

Developer Tools: Git, Linux, Docker, Kubernetes, Jira

PROJECTS

Khabo-Koi — Restaurant Discovery & Table Reservation Platform

Technologies: React.js, Vite, Tailwind CSS, Django REST Framework, PostgreSQL | Live Demo | GitHub

- Restaurant discovery, table availability and transaction-safe reservations across 51 REST API routes.
- Food pre-orders, AI dining assistant and four role-based dashboards; backend includes 192 automated test methods.

CampusLink — University Community Platform

Technologies: React.js, Vite, Tailwind CSS, Spring Boot, REST APIs, PostgreSQL | Live Demo | GitHub

- Community platform with 62 HTTP handlers across lost-and-found, borrowing, complaints and talent sharing.
- Spring Boot backend with 15 JPA entities, JWT role-based access, moderation, notifications and async activity logging.

RESEARCH PAPERS & PUBLICATIONS

Workload-Based Energy Efficiency Analysis of Native and Virtualized Operating Systems for Sustainable Computing: An Experimental Study on Modern Computing Platforms

Submitted — IEEE Computer Society Bangladesh Chapter Symposium 2026

- Compared native macOS with Windows 11 ARM and Ubuntu 24.04 ARM virtual machines on Apple M3 using four workloads, each measured in five 600-second runs with EnergiBridge.
- Analyzed average power, energy, 95% confidence intervals and carbon estimates; macOS performed best in video/web tasks, while Windows VM was more efficient for idle/Python tasks.

RESEARCH PAPERS & PUBLICATIONS (CONTINUED)

Colorism in Vision-Language Models: A Bangladeshi Study of Skin-Tone and Gender Bias in VLMs

Undergraduate Thesis | Manuscript in Preparation

- Evaluated four vision-language models using 100 synthetic Bangladeshi facial images (50 paired identities), 1,200 socioeconomic profiles and 4,000 simulated recruitment decisions.
- Found Gemini's light-skin profiles received BDT 8,877 higher estimated income ($p < 0.001$); light-skin selection rates were 66.87% (Gemini), 66.35% (GPT) and 59.76% (DeepSeek).
- Observed candidate-position confounding in Llama: Candidate 1 was selected in all 1,000 fixed-seed recruitment trials.

Robustness and Enhancement Recovery of CNN and Vision Transformer Models for Rice Leaf Disease Classification Under Controlled Image Degradation

Manuscript in Preparation

- Benchmarked MobileNetV3-Small (71.08% accuracy) and DeiT-Tiny (74.51%) on seven-class rice leaf disease images under noise, blur and low-light degradation.
- Evaluated gamma correction, CLAHE and unsharp masking using accuracy, macro-F1, PSNR and SSIM; gamma raised MobileNetV3-Small low-light accuracy by 8.33 percentage points.

Empirical Evaluation of Living-off-the-Land (LotL) Attack Techniques and Security Control Limitations in Modern Operating Systems

Manuscript in Preparation

- Investigated contextual detection of Living-off-the-Land techniques across Windows and macOS using a catalog of 40 techniques and 200 synthetically constructed evaluation traces.
- Developed and evaluated two rule-based detection approaches using native-tool activity and contextual features, including execution sequences, timing, process relationships, network activity, and persistence behavior, across Windows and macOS.

HONORS & AWARDS

Academic Merit Scholarship — Spring 2026

United International University, Dhaka, Bangladesh

- Awarded in recognition of academic performance during the Spring 2026 semester.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Joint Secretary (Administration) — 13th Executive Committee

UIU Debate Club, United International University

- Served as Joint Secretary (Administration), contributing to the club's administrative and organizational activities.
- Participated in two national-level debate competitions.